How to Use the LaTeX Project Template

A Step-by-Step Guide for Students

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Goal of this Tutorial

This short guide shows you how to use the provided template to: (1) set your title and names, (2) add figures and tables, (3) cite references with `natbib`, (4) use acronyms with `glossaries`, and (5) compile successfully. If you follow the exact folder names and commands here, your project will compile without issues.

1 Folder Structure (Recommended)

Create the following structure (exact names):

```
ProjectRoot/
|-- main.tex
|-- acronyms/
|   `-- acronyms.tex
|-- Chapters/
|   |-- Chapter_1.tex
|   |-- Chapter_2.tex
|   |-- Chapter_3.tex
|   |-- Chapter_4.tex
|   `-- Chapter_5.tex
|-- Figures/
|   `-- Logo_UoS.png   (or any figures you add)
`-- Bibliography/
    `-- ref.bib
```

2 Your Main File

Use the provided `main.tex` from the template. The key parts are shown here so you know what to edit:

`main.tex` (key excerpts)

```
\documentclass[12pt]{report}
% ... (all packages from the template)

% Load acronyms file early (before hyperref)
\input{./acronyms/acronyms}
```

```
% Load hyperref LAST
\usepackage[hidelinks]{hyperref}

\begin{document}

% ---- Title Page ----
\begin{titlepage}
  \centering
  \includegraphics[width=0.25\textwidth]{Figures/Logo_UoS.png}

  \textsc{\Large College of Computing and Informatics\\
  Department of Computer Engineering}\\[1.5cm]

  \textsc{\Large SPD Proposal}\\[0.5cm]

  \rule{\linewidth}{0.5mm} \\[0.4cm]
  {\LARGE \bfseries Zero Trust Device to Device Communication}\\[0.4cm]
  \rule{\linewidth}{0.5mm} \\[1.5cm]

  \begin{flushleft}
  \textbf{Students: } Ali Ahmed, Sara Omar \\
  \textbf{Supervisor: Dr. Kais Belwafi} \\
  \textbf{Term:} Fall 2025--2026
  \end{flushleft}

  \vfill
  \textbf{Date:} \today
\end{titlepage}

\tableofcontents
\listoffigures
\listoftables
\printglossary[type=\acronymtype, title=Acronyms]

\chapter*{Abstract}
\addcontentsline{toc}{chapter}{Abstract}
% Abstract text here...

\chapter{Introduction}
\input{Chapters/Chapter_1}

\chapter{Literature Review}
\input{Chapters/Chapter_2}

\chapter{Technical Background}
```

```
\input{Chapters/Chapter_3}

\chapter{Proposed Solution}
\input{Chapters/Chapter_4}

\chapter{Research Methodology}
\input{Chapters/Chapter_5}

\bibliographystyle{plainnat}
\bibliography{Bibliography/ref}
\end{document}
```

Edit the Title/Names In the title page block, replace the bold line with *your* project title, and update the **Students:** line with the correct names.

3 Chapters: Where to Write

Each chapter lives in `Chapters/Chapter_X.tex`. Start with `Chapter_1.tex`. Example content:

Chapters/Chapter_1.tex

```
In this chapter, we introduce the problem and define the scope.
We will reference acronyms like \gls{bci} and \gls{uav}.

\section{A Simple TikZ Figure}
Figure~\ref{fig:block} shows a minimal pipeline.

\begin{figure}[h!]
  \centering
  \begin{tikzpicture}[
    node distance=2.2cm,
    block/.style={rounded corners, draw, align=center, minimum width=3cm,
    minimum height=1cm}
  ]
    \node[block] (s) {Sensors};
    \node[block, right=of s] (p) {Preprocess};
    \node[block, right=of p] (c) {Classifier};
    \draw[->] (s) -- (p);
    \draw[->] (p) -- (c);
  \end{tikzpicture}
  \caption{High-level processing pipeline.}
  \label{fig:block}
\end{figure}

\section{A Clean Table}
See Table~\ref{tab:ex} for formatting with booktabs.

\begin{table}[h!]
\centering
\caption{Example performance summary}
\label{tab:ex}
\begin{tabular}{lcc}
\toprule
\textbf{Algorithm} & \textbf{Accuracy (\%)} & \textbf{Latency (ms)} \\
\midrule
CSP      & 85.2 & 18 \\
FBCSP    & 87.6 & 24 \\
XAI-CSP  & 90.1 & 30 \\
\bottomrule
\end{tabular}
\end{table}

\section{Citing References}
Wearable EEG systems are surveyed by \citet{casson2019wearable};
a broader algorithm review appears in \citep{lotte2018review}.
```

4 Figures: What to Do

1. Put image files (PNG/PDF/JPG) in the `Figures/` folder.
2. Use `\includegraphics` to insert them.

Insert a PNG figure

```
\begin{figure}[h!]
  \centering
  \includegraphics[width=0.6\textwidth]{Figures/your_figure.png}
  \caption{System architecture of the proposed method.}
  \label{fig:arch}
\end{figure}
As shown in Figure~\ref{fig:arch}, ...
```

5 Tables: Best Practices

Use `booktabs` for clean lines and avoid vertical rules. See the example in Chapter 1. Always add a `\caption` and a `\label` so you can reference the table in your text.

6 References with `natbib`

Add entries to `Bibliography/ref.bib`

`Bibliography/ref.bib` (sample entries)

```
@article{casson2019wearable,
  title   = {Wearable EEG and beyond},
  author  = {Casson, Alexander J.},
  journal = {Biomedical Engineering Letters},
  year    = {2019},
  volume  = {9},
  number  = {1},
  pages   = {53--71},
  publisher = {Springer}
}

@article{lotte2018review,
  title   = {A review of classification algorithms for EEG-based brain--computer
     interfaces: a 10-year update},
  author  = {Lotte, Fabien and Bougrain, Laurent and Cichocki, Andrzej and Clerc
    , Maureen and Congedo, Marco and Rakotomamonjy, Alain and Yger, Florian},
  journal = {Journal of Neural Engineering},
  year    = {2018},
  volume  = {15},
  number  = {3},
  pages   = {031005}
}
```

Cite inside chapters

Citation examples

```
Textual citation: \citet{casson2019wearable} report that...
Parenthetical:   ...as shown previously \citep{lotte2018review}.
```

Print the bibliography

Already handled at the end of `main.tex`:

```
\bibliographystyle{plainnat}
\bibliography{Bibliography/ref}
```

7 Acronyms with `glossaries`

Define them in `acronyms/acronyms.tex`

acronyms/acronyms.tex

```
\newacronym{bci}{BCI}{Brain-Computer Interface}
\newacronym{uav}{UAV}{Unmanned Aerial Vehicle}
\newacronym{iot}{IoT}{Internet of Things}
\newacronym{tls}{TLS}{Transport Layer Security}
\newacronym{fpga}{FPGA}{Field-Programmable Gate Array}
```

Use them in your text

```
We develop a \gls{bci} for \gls{uav} control over \gls{tls} links on \gls{fpga}.
```

Print the list

Already present in `main.tex`:

```
\printglossary[type=\acronymtype, title=Acronyms]
```

8 How to Compile (Exact Commands)

From `ProjectRoot/` in a terminal:

Option A: Traditional (BibTeX + Glossaries)

```
pdflatex main
bibtex main
makeglossaries main
pdflatex main
pdflatex main
```

Option B: latexmk convenience (plus glossaries)

```
latexmk -pdf main.tex
makeglossaries main
latexmk -pdf main.tex
```

9 Common Errors & Quick Fixes

- **"Environment subfigure undefined"**
  You used `subcaption`; the correct environment is `subfigure` from `subcaption`?
  Use:

```
\begin{figure}
  \centering
  \begin{subfigure}{0.48\linewidth}
    \includegraphics[width=\linewidth]{Figures/a.png}
    \caption{A}
  \end{subfigure}
  \begin{subfigure}{0.48\linewidth}
    \includegraphics[width=\linewidth]{Figures/b.png}
    \caption{B}
  \end{subfigure}
  \caption{Two panels.}
\end{figure}
```

Ensure `\usepackage{subcaption}` is loaded (already in the template).

- **"??" in references or missing entries**
  Run the full compile sequence (pdflatex → bibtex → makeglossaries → pdflatex ×2). Check the path and name of `Bibliography/ref.bib` and `\bibliography{Bibliography/ref}`.
- **Acronyms do not print / appear as keys**
  Make sure `\makeglossaries` is in the preamble, the acronyms file is loaded *before* `hyperref` (`\input{./acronyms/acronyms}`), and you ran `makeglossaries main` once after BibTeX.
- **Figure not found**
  Check the file name and path (case-sensitive), e.g., `Figures/MyPlot.png`, and that you used the correct extension.
- **Undefined control sequence `\hline`**
  Usually caused by a broken `tabular` environment (missing `\begin{tabular}...` or mismatched `&` or `\\`). Verify your table block is complete.

10 Quality Tips

- Keep figures in vector format (PDF) when possible; set width relative to `\textwidth`.
- Use concise, informative captions; reference every figure/table in the text.
- Prefer `booktabs` for tables and avoid vertical rules.
- Keep acronyms consistent and define them once.
- Commit to a bibliography style (e.g., `plainnat`) and stick to it.

You Are Ready

Use this tutorial alongside the template. Update the title page, start writing in `Chapters/`, add your references to `Bibliography/ref.bib`, and compile using the commands above. If you get stuck, re-check the paths and the compile order.

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